

Advanced (Habanero)

- Q1.** What is the unit rate of $12 \div 4$?
(A) 2
(B) 3
(C) 4
(D) 6
(E) 8
- Q2.** If y varies directly with x and $y = 12$ when $x = 4$, what is the constant of variation?
(A) 2
(B) 3
(C) 4
(D) 6
(E) 8
- Q3.** A car travels 250 miles in 5 hours. What is its speed in miles per hour?
(A) 25 mph
(B) 50 mph
(C) 50 mph
(D) 75 mph
(E) 100 mph
- Q4.** If z varies inversely with w and $z = 16$ when $w = 4$, what is z when $w = 2$?
(A) 2
(B) 4
(C) 8
(D) 16
(E) 32
- Q5.** What is the unit rate of $36 \div 6$?
(A) 1
(B) 2
(C) 4
(D) 6
(E) 8
- Q6.** If x and y vary directly and $y = 20$ when $x = 5$, what is y when $x = 3$?
(A) 8
(B) 10
(C) 12
(D) 15
(E) 20
- Q7.** If y varies inversely with the square of x and $y = 4$ when $x = 2$, what is y when $x = 4$?
(A) 0.25
(B) 0.5
(C) 1
(D) 2
(E) 4
- Q8.** A cyclist travels 25 miles in 2 hours. What is the unit rate?
(A) 10 mph
(B) 12.5 mph
(C) 15 mph
(D) 20 mph
(E) 25 mph

Open Ended Questions (FRQ)

- Q9.** A water tank can be filled by two pipes. Pipe A can fill the tank in 6 hours and pipe B can fill it in 4 hours. How long will it take to fill the tank if both pipes are used together?
- Q10.** The cost of manufacturing x units of a product is given by the function $C(x) = 2x^2 + 3x + 1000$. Find the rate of change of cost with respect to the number of units produced when $x = 10$.

Chef's Tips

- Read each question carefully before answering
- Show all work for free response questions
- Use a calculator only if necessary